

Fresh Trace

APPENDIX A (DESIGN DOCUMENTS)

Submitted by:

Team 5:

Sinikiem Azaiki

Burak Koseoglu

Rexy Sara Raju

Krish Mukesh Singh

Ba Thinh Vuong

Amirhossein Mohammadi

Submitted to: Nesovic, Jake

ICET Department,

School of Engineering Technology and Applied Science

Progress Campus, Block A

Centennial College

Discipline: Software Engineering

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A1. System Architecture Diagram (Developer's perspective):

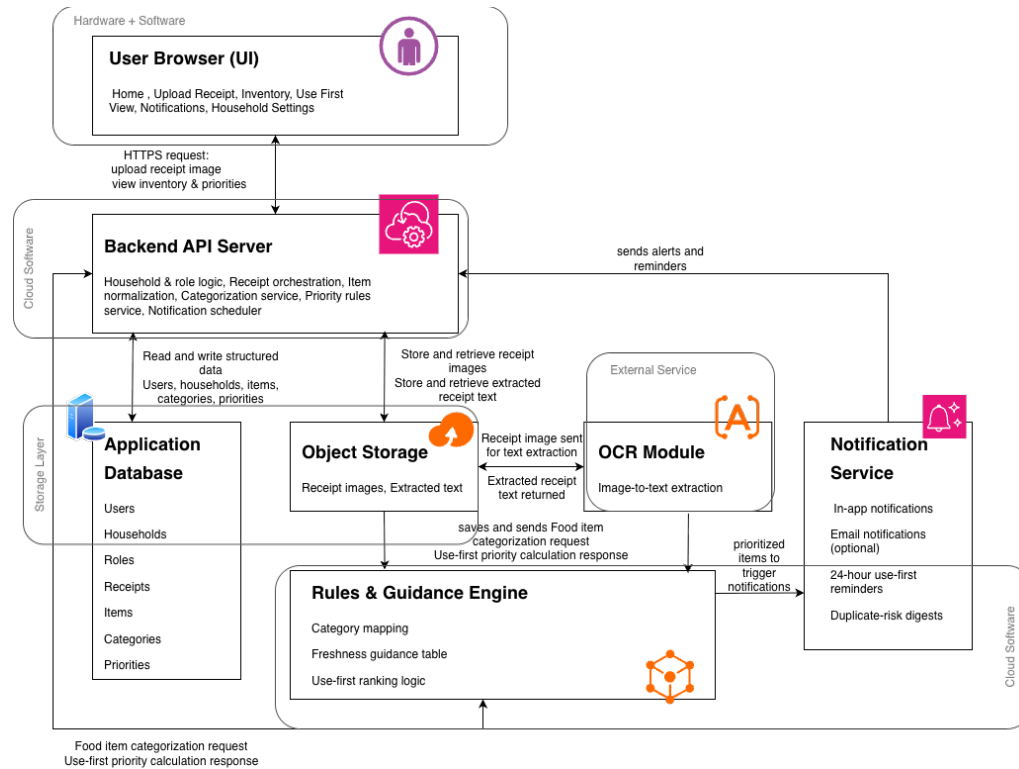


Fig 1: System Architecture Diagram

A2. Use Case Diagram

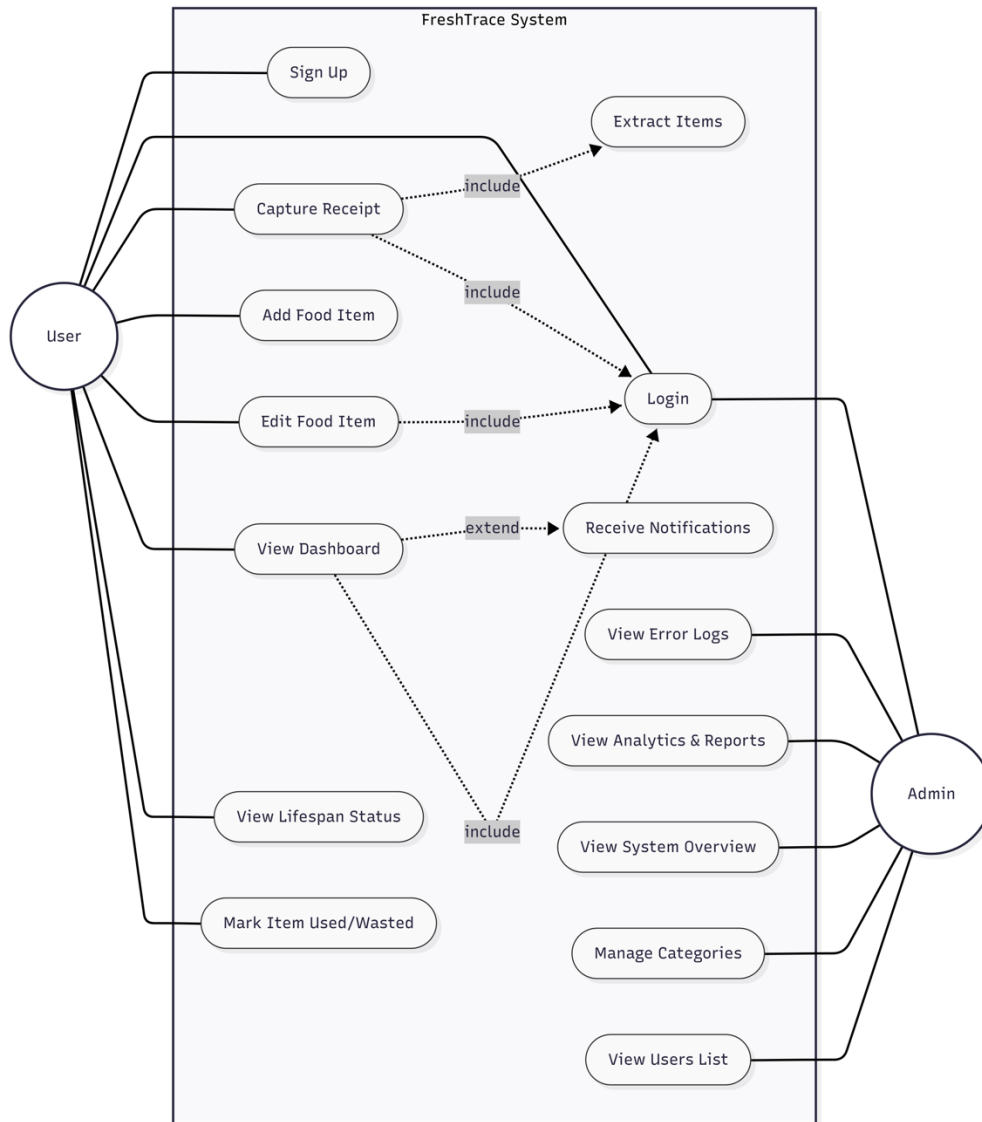


Fig 2: Use Case Diagram

A3. Class Diagram

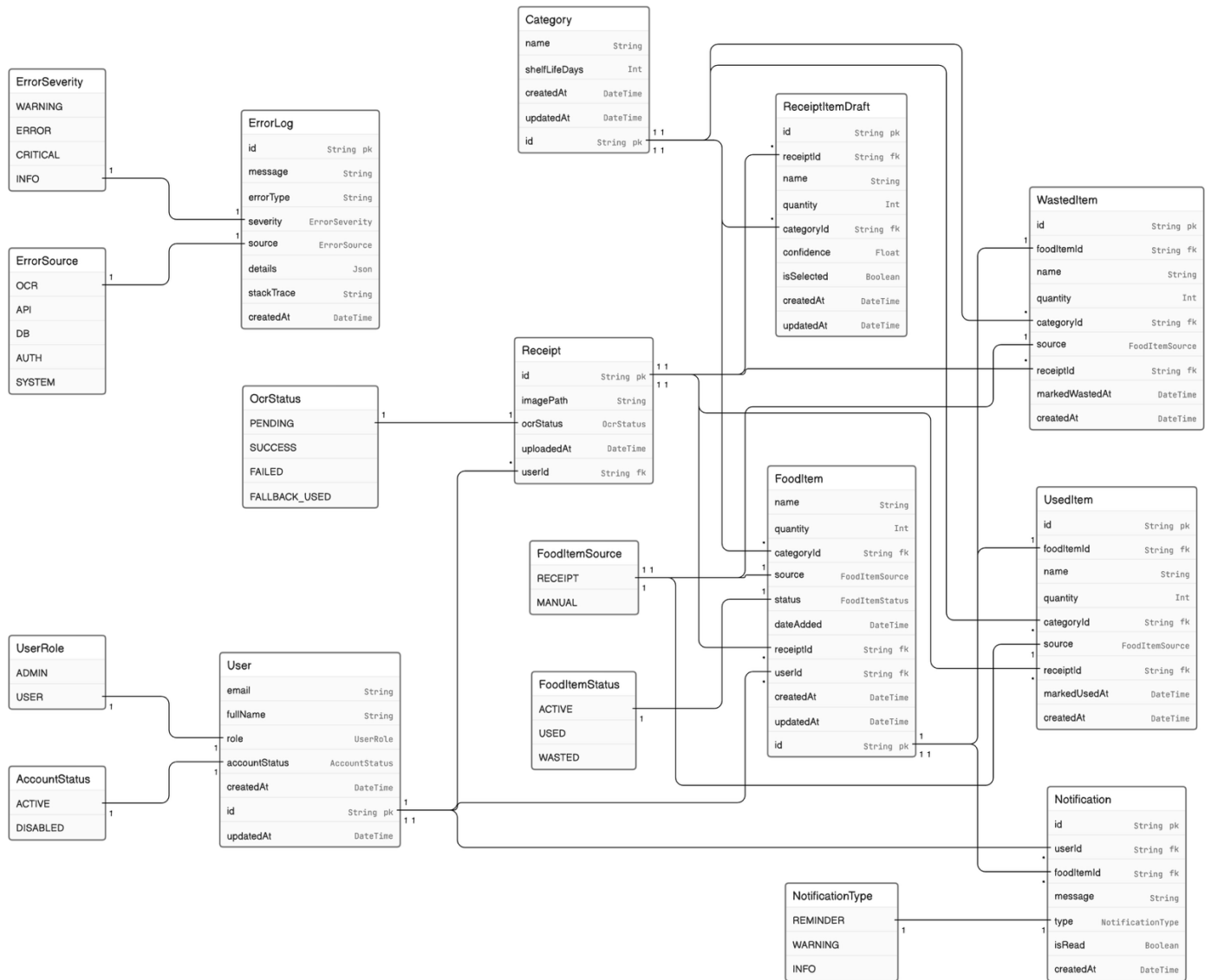


Fig 3: Class Diagram

A4. Object Diagram

Object Diagram – FreshTrace Food Inventory State Change

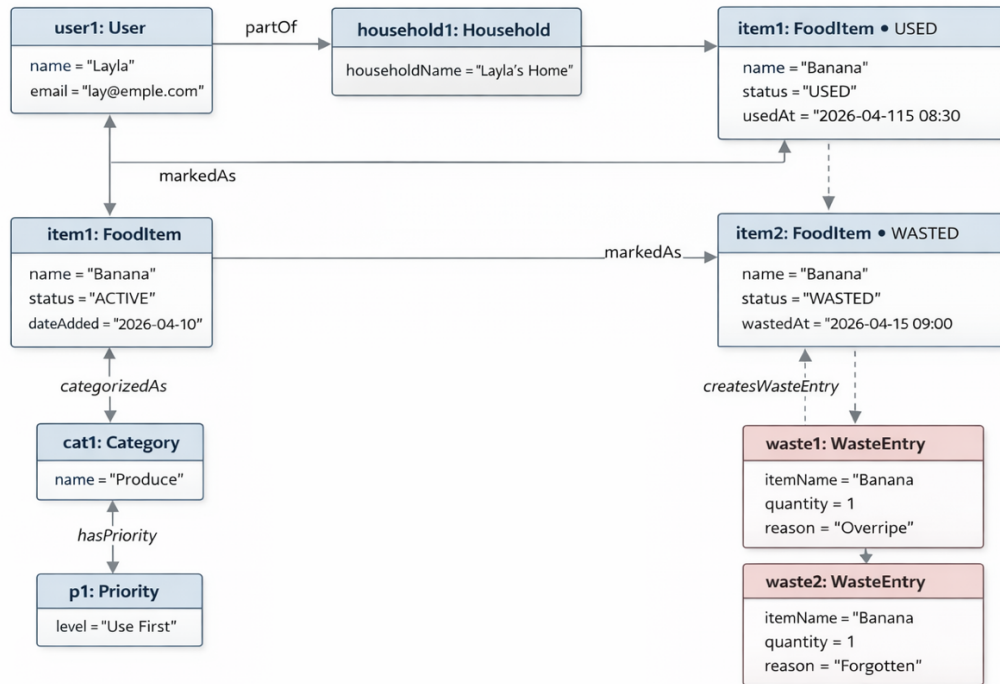


Fig 4: Object Diagram 1 (Inventory state change)

Object Diagram – FreshTrace Food Expiry Notification

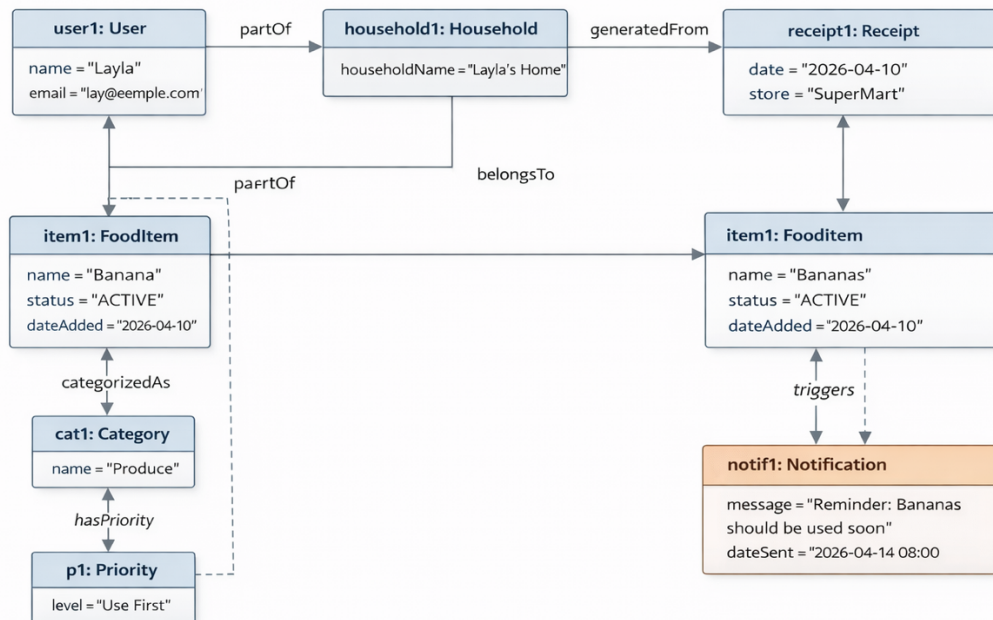


Fig 5: Object Diagram 2 (Food expiry notification)

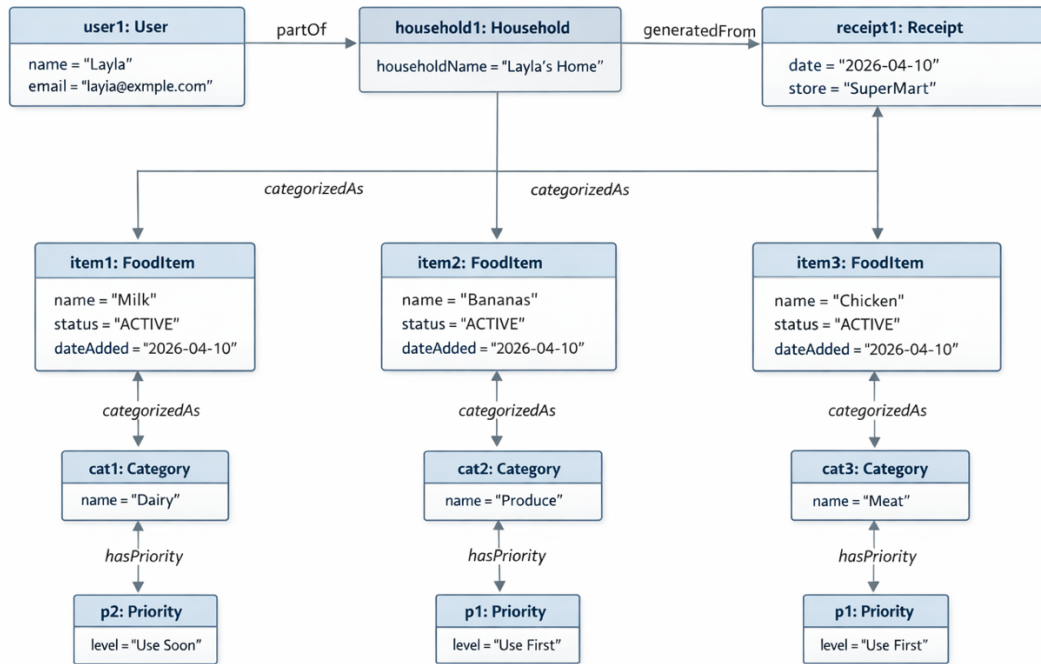


Fig 6: Object Diagram 3 (Grocery tracking system)

A5. Component Diagram

Component Diagram – FreshTrace Receipt Processing Flow

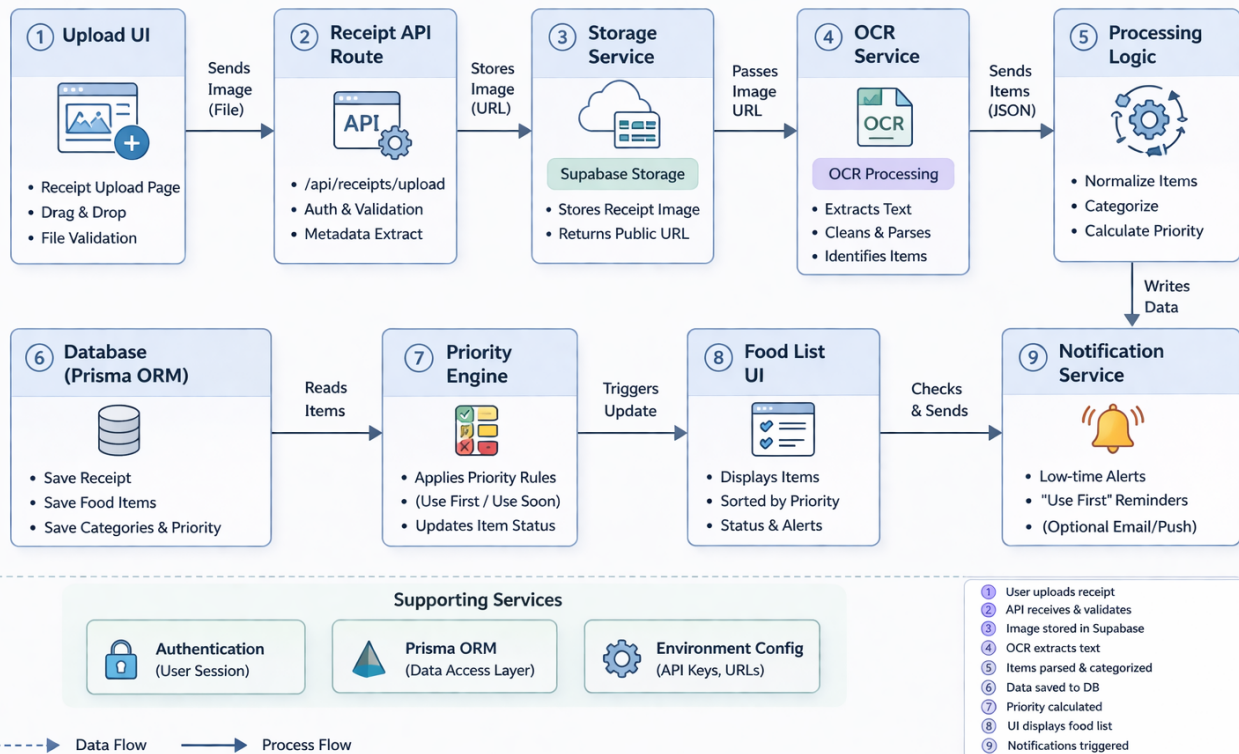


Fig 7: Component Diagram

A6. Deployment Diagram

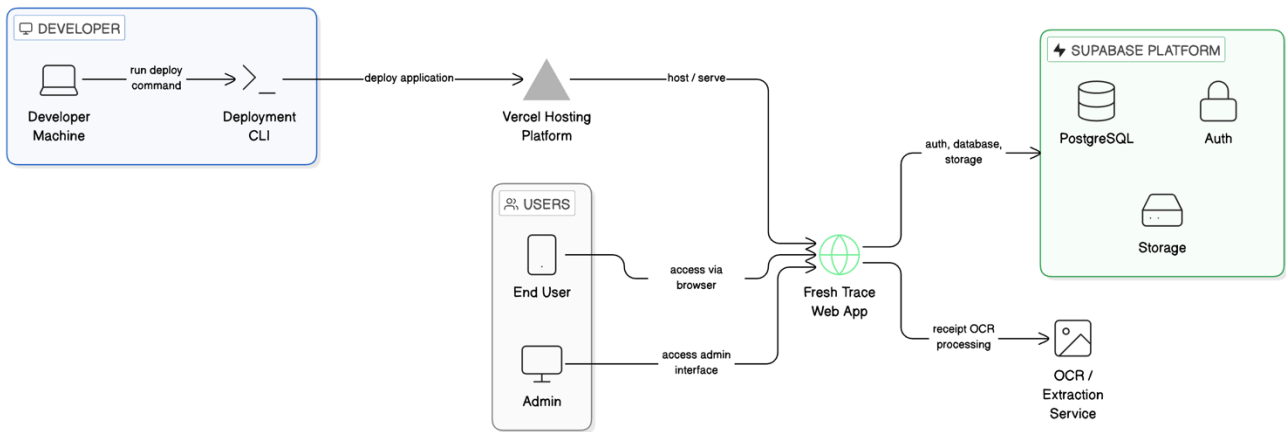


Fig 8: Deployment Diagram

A7. Activity Diagram

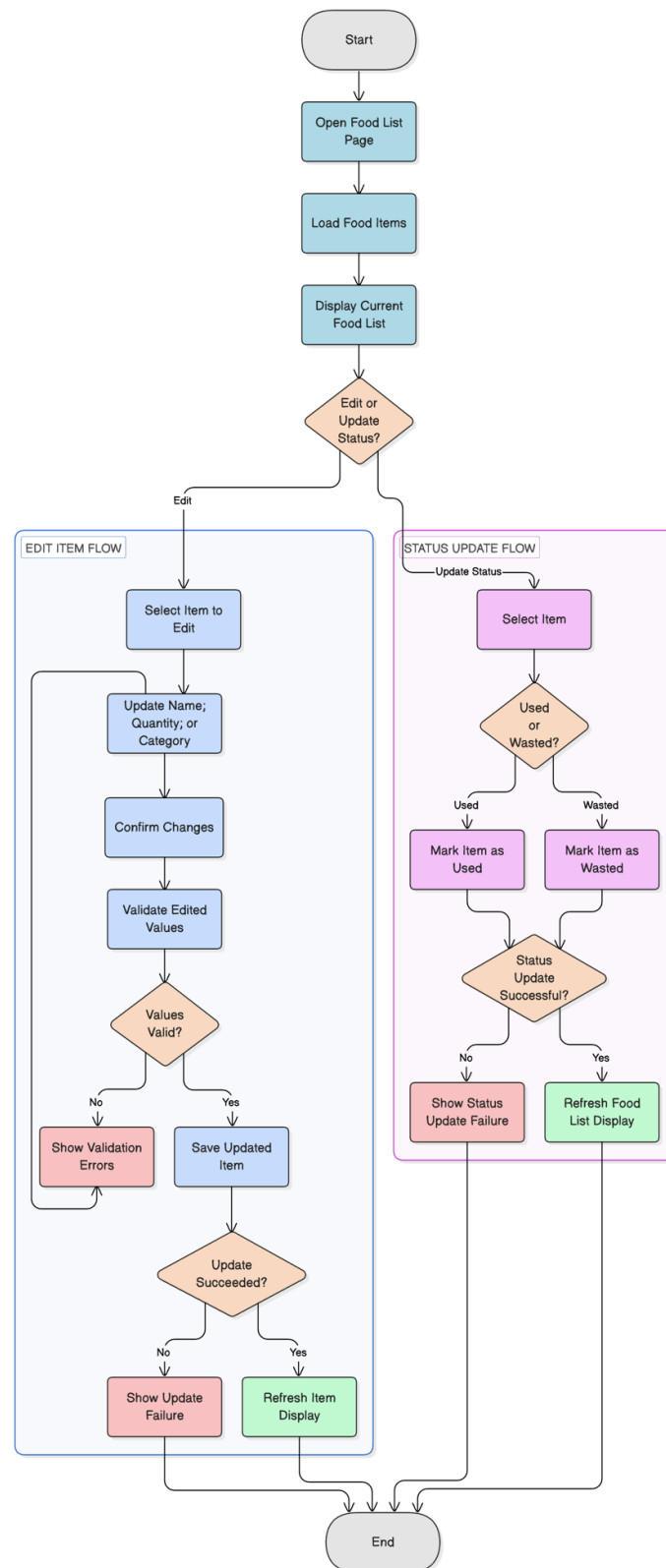


Fig 9: Activity Diagram (Users Food list)

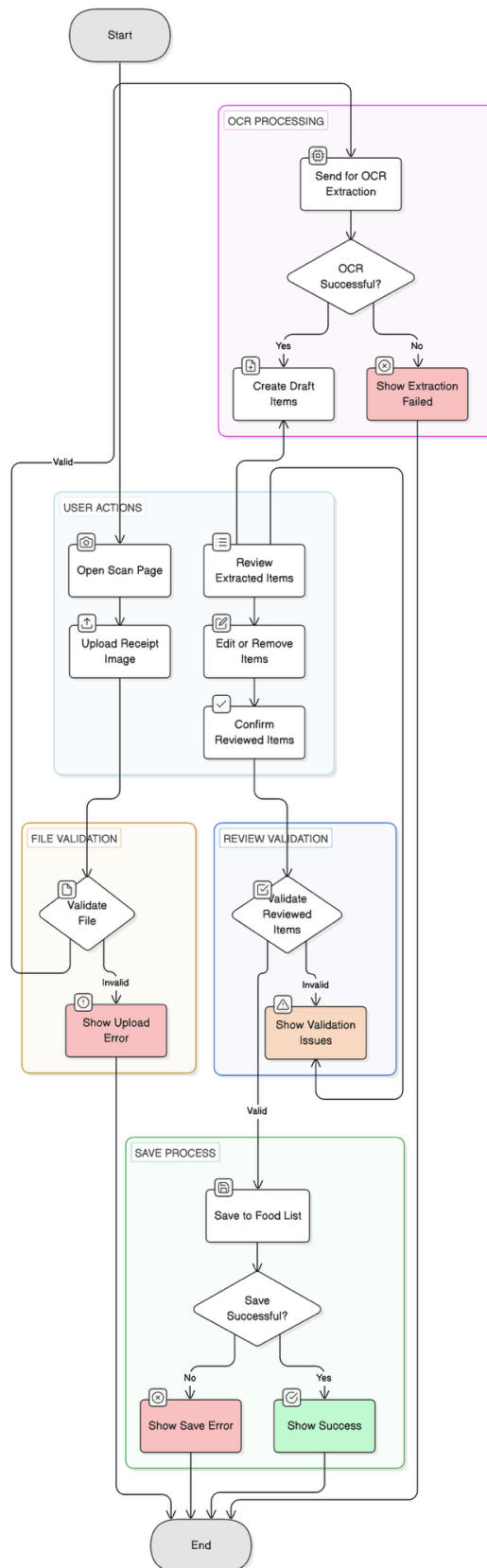


Fig 10: Activity Diagram (Users Scan Receipt)

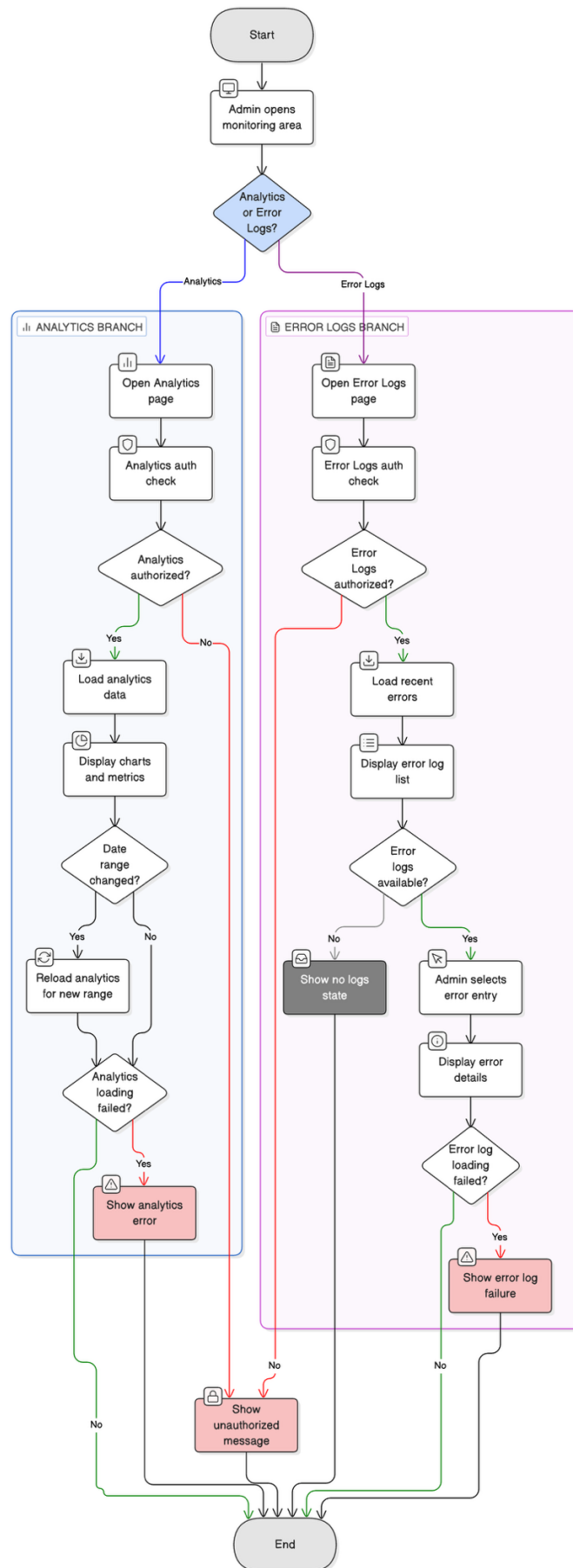


Fig 11: Activity Diagram (Analytics and Error logs)

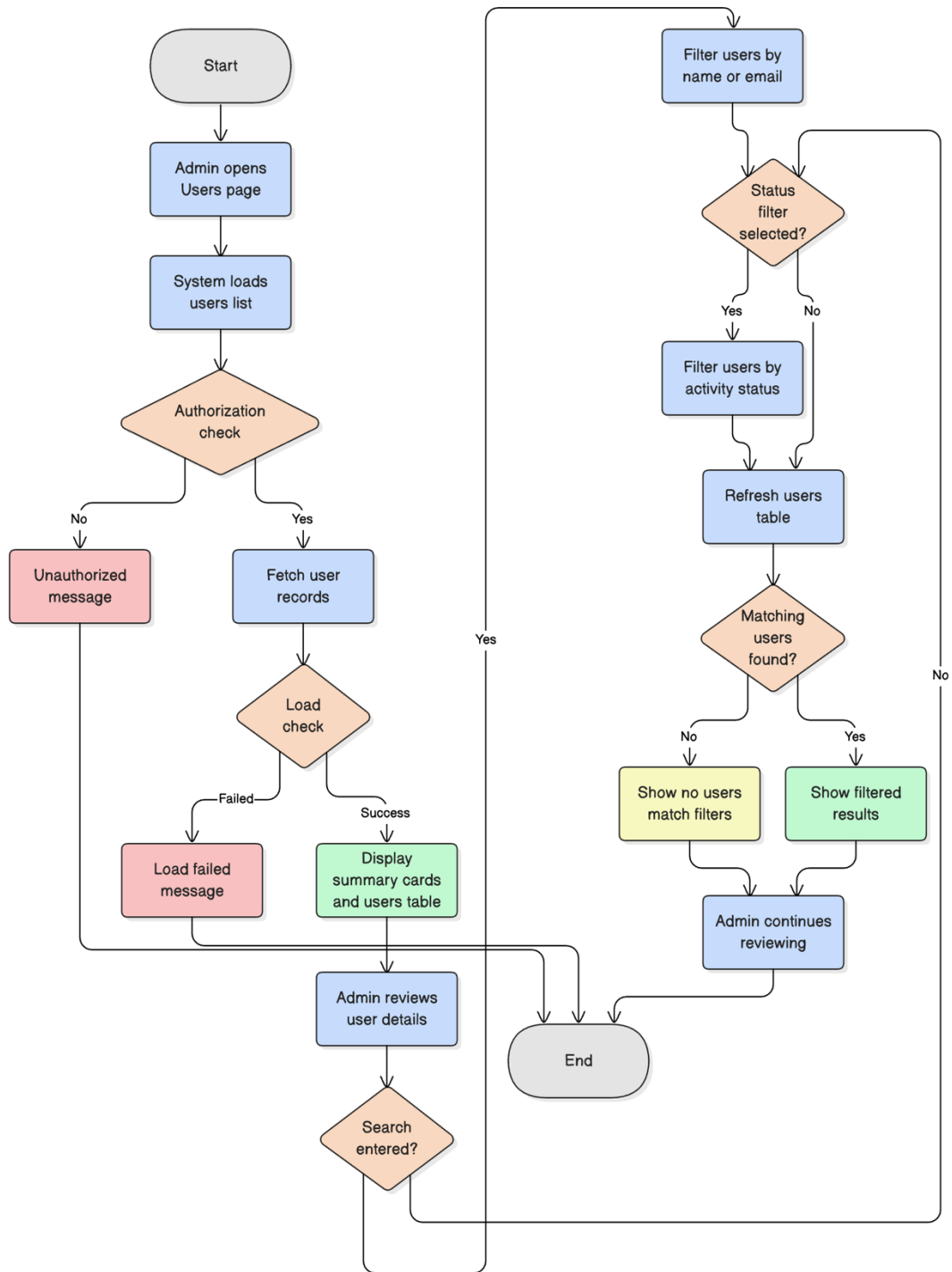


Fig 12: Activity Diagram (Admin Users List)

A8. Sequence Diagram

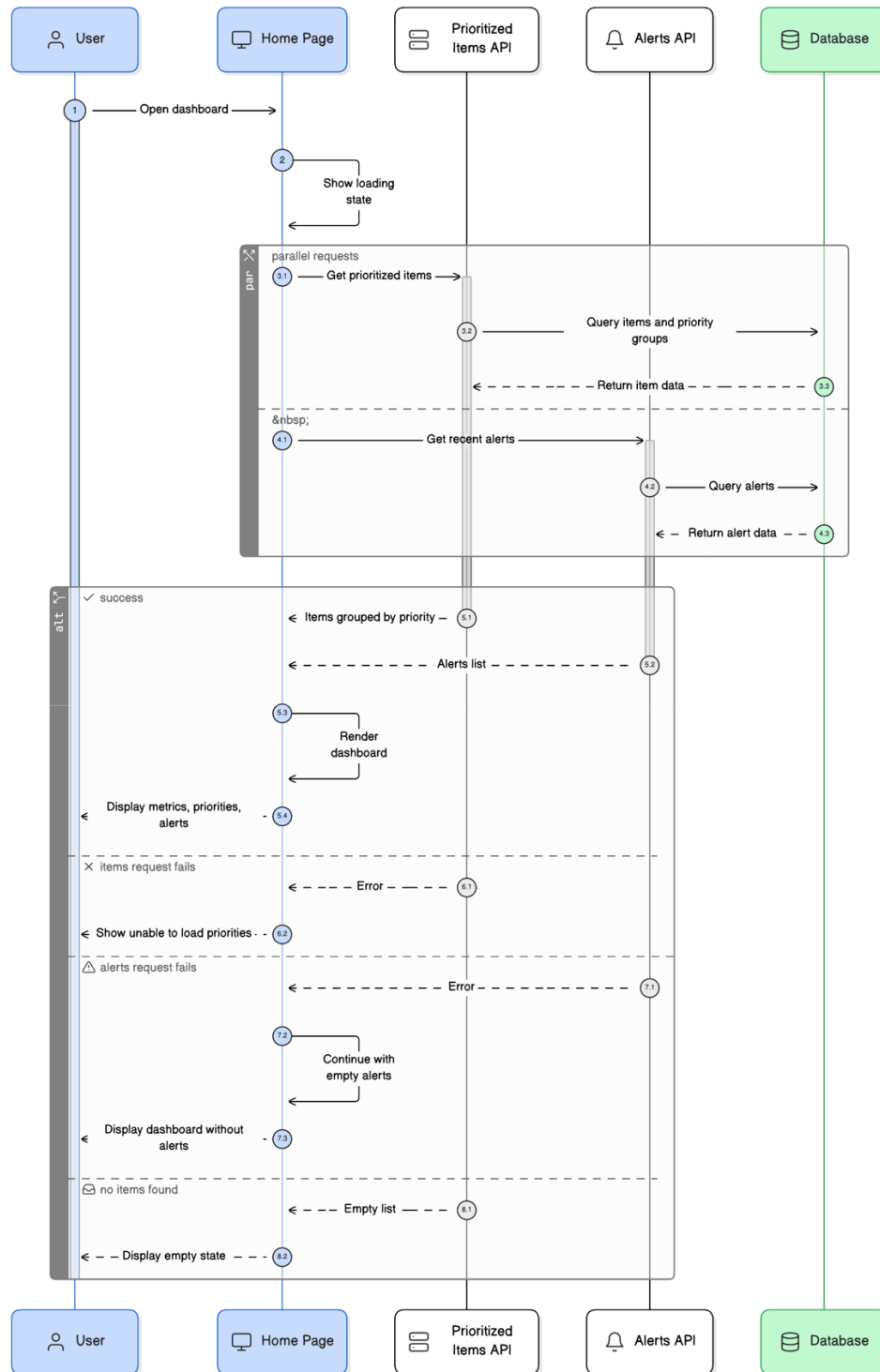


Fig 13: Sequence Diagram (User dashboard)

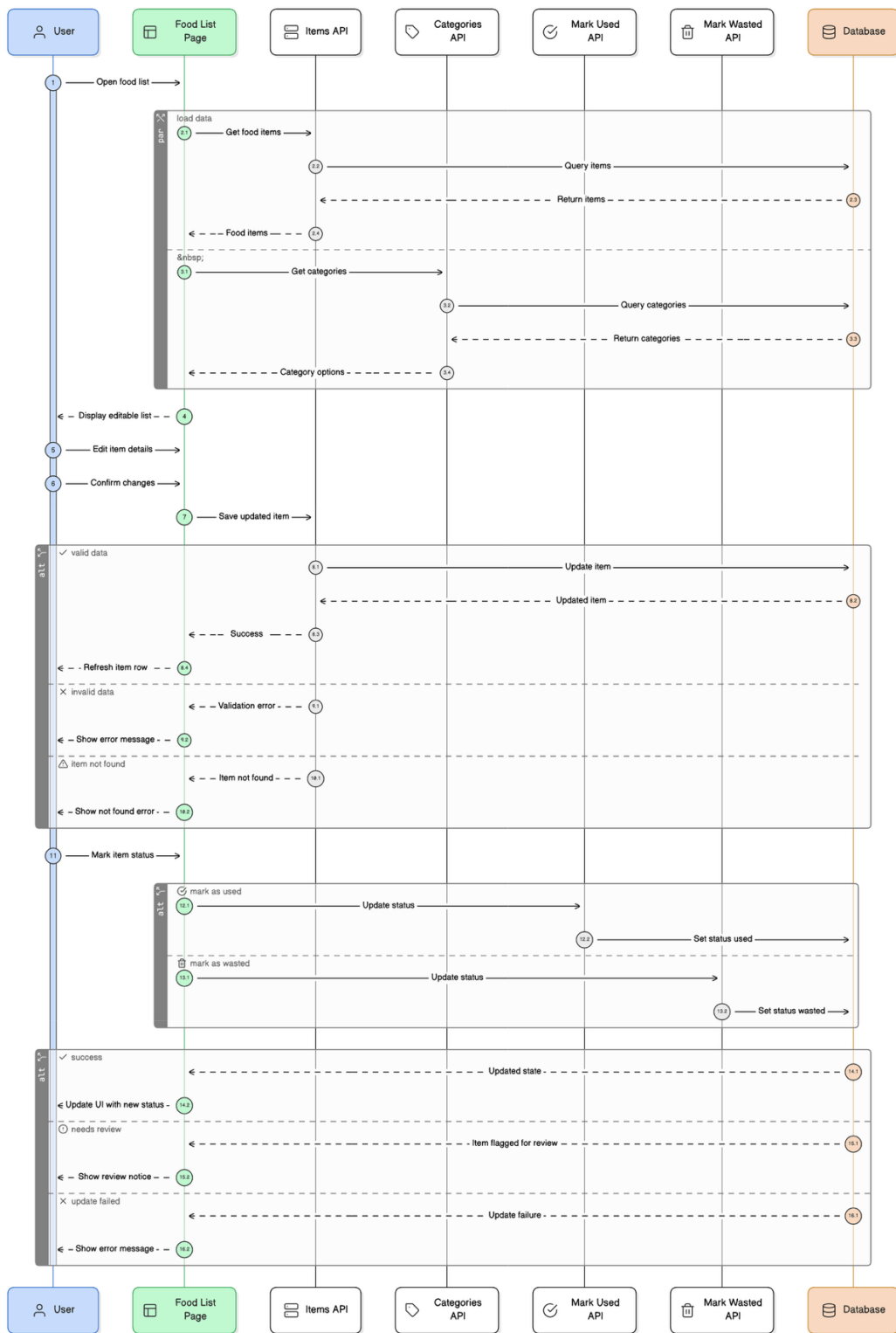


Fig 14: Sequence Diagram (User food list)

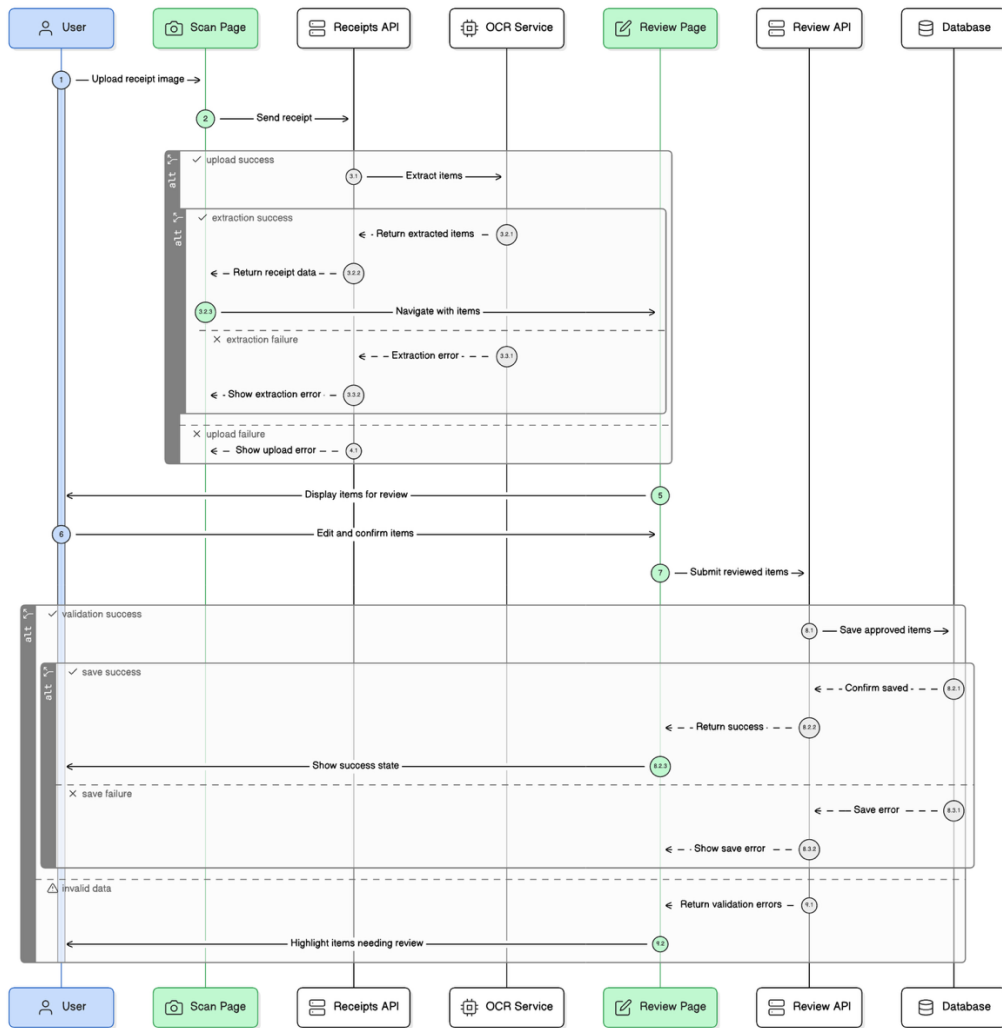


Fig 15: Sequence Diagram (User scan receipt)

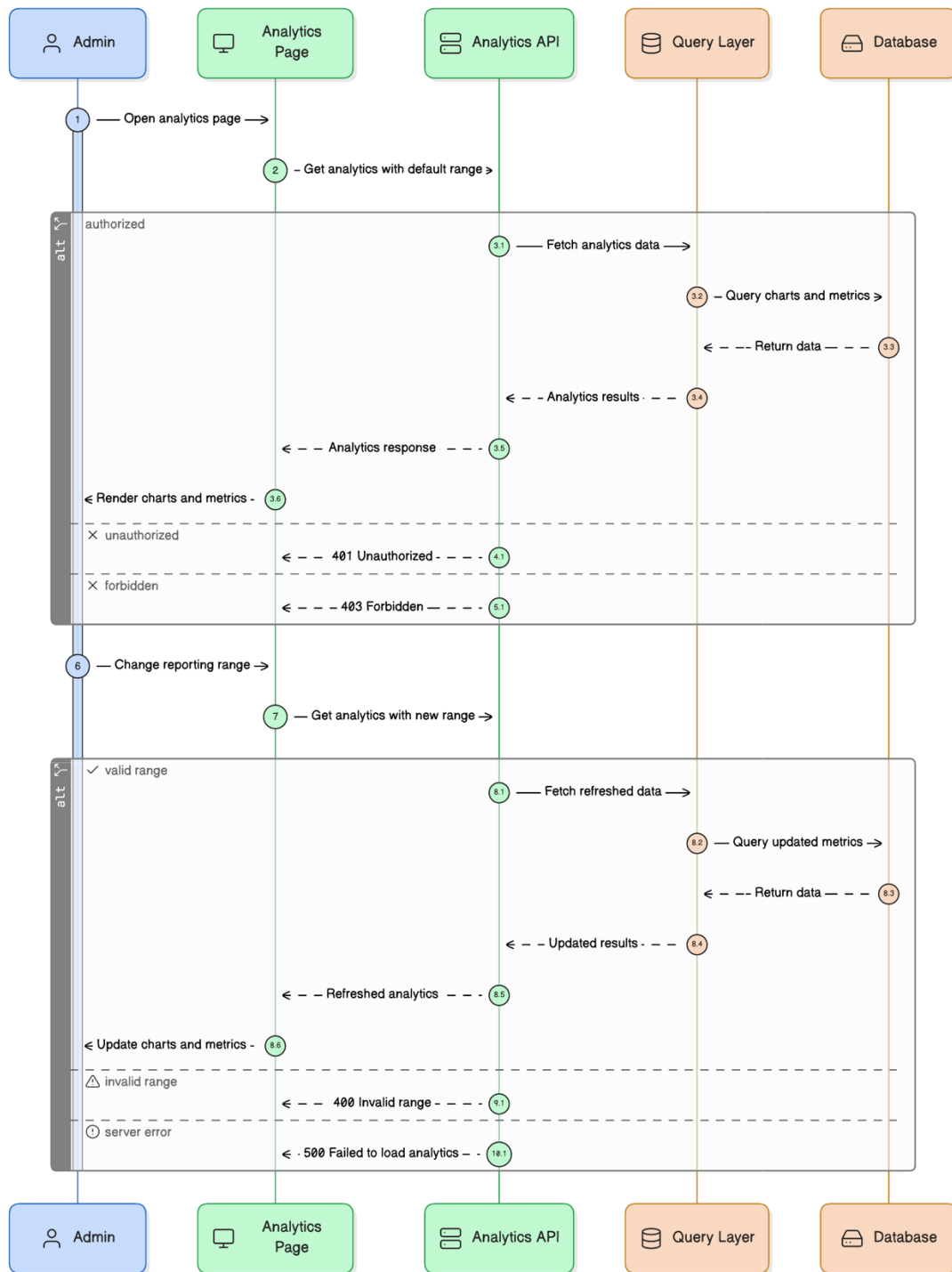


Fig 16: Sequence Diagram (Admin analytics)

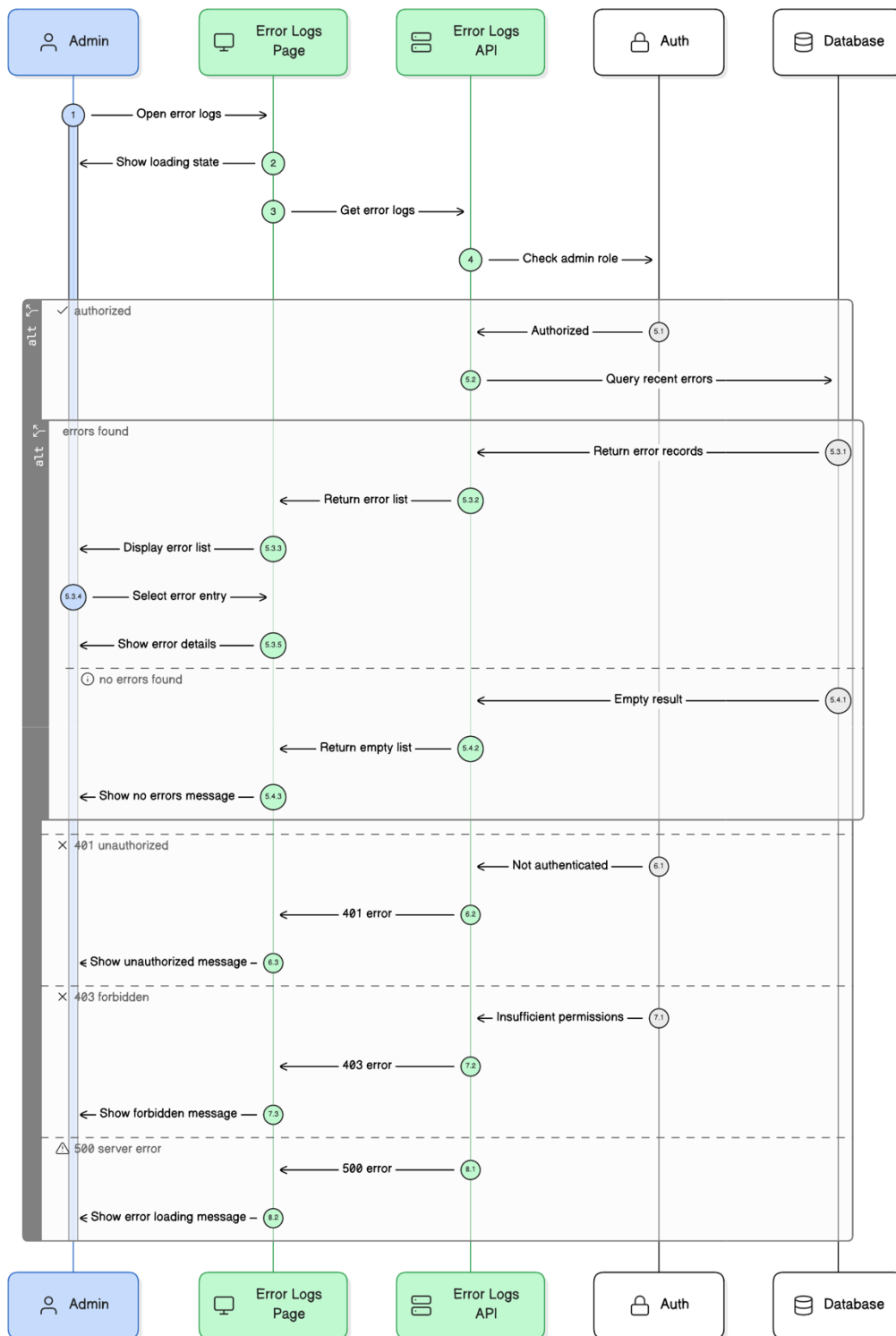


Fig 17: Sequence Diagram (Admin error logs)

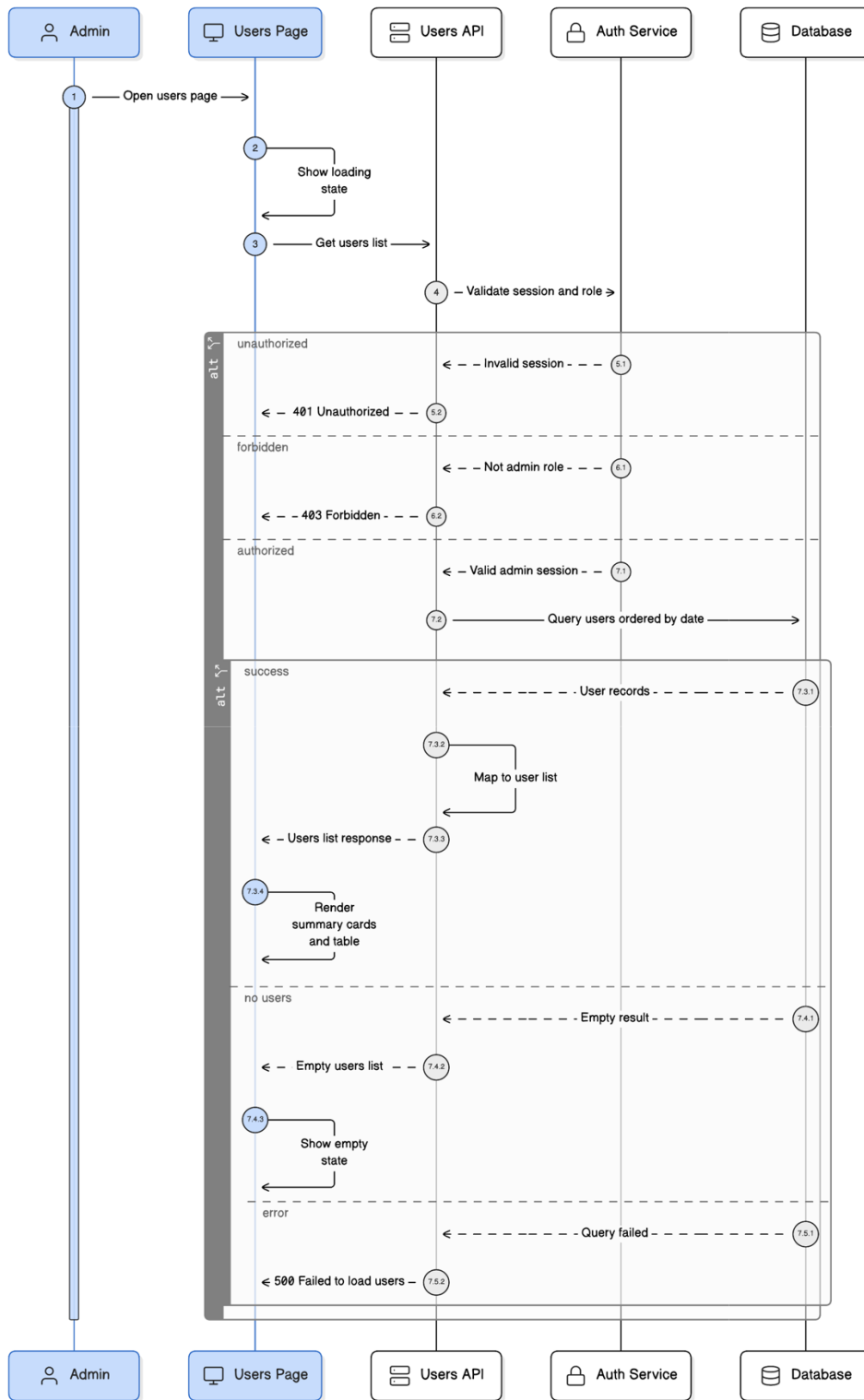


Fig 18: Sequence Diagram (Admin users list)

A9. ERD Diagram

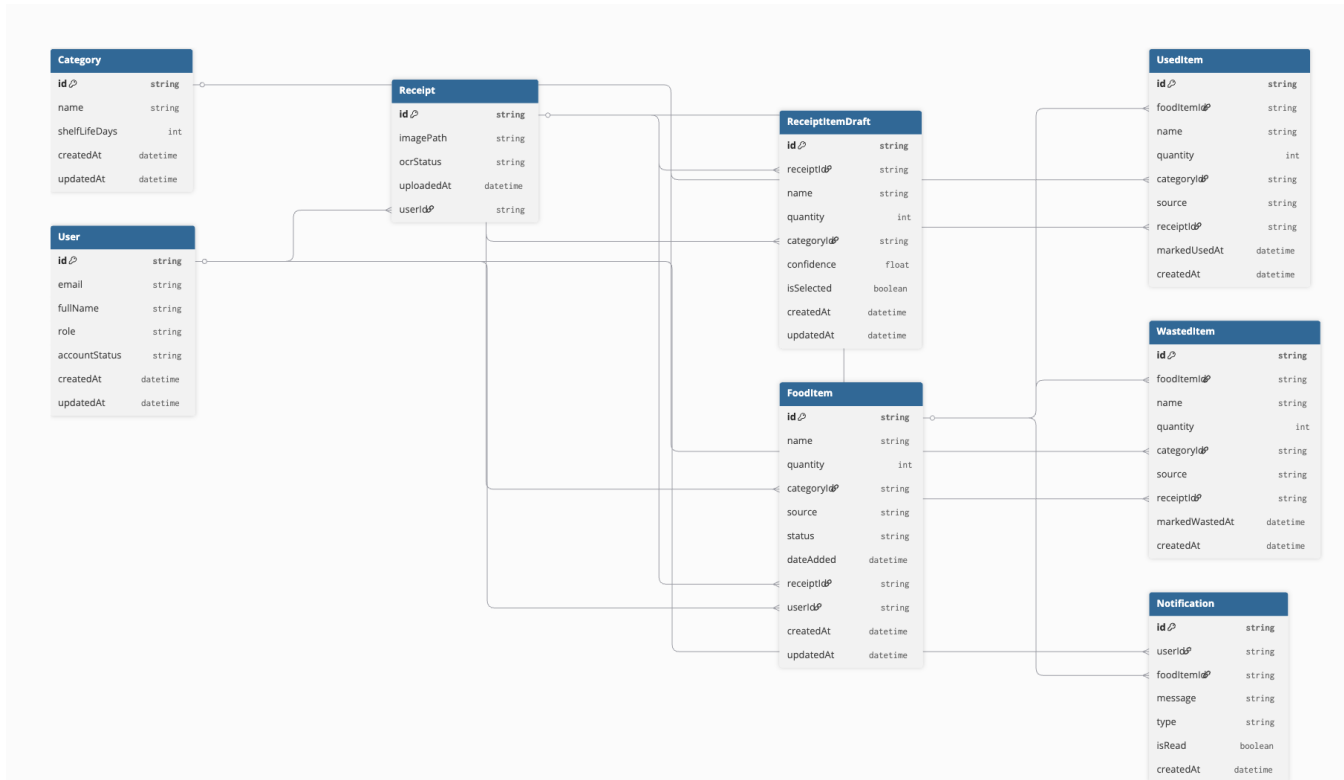


Fig 19: ERD Diagram

A10. Paper Prototype of Fresh Trace

As an Active/Occasional user, I want to see which food items should be used first so that I can decide what to cook or eat next (&) As an Active/Occasional user, I want to view the lifespan status of food items so that I can understand how long they have been stored.

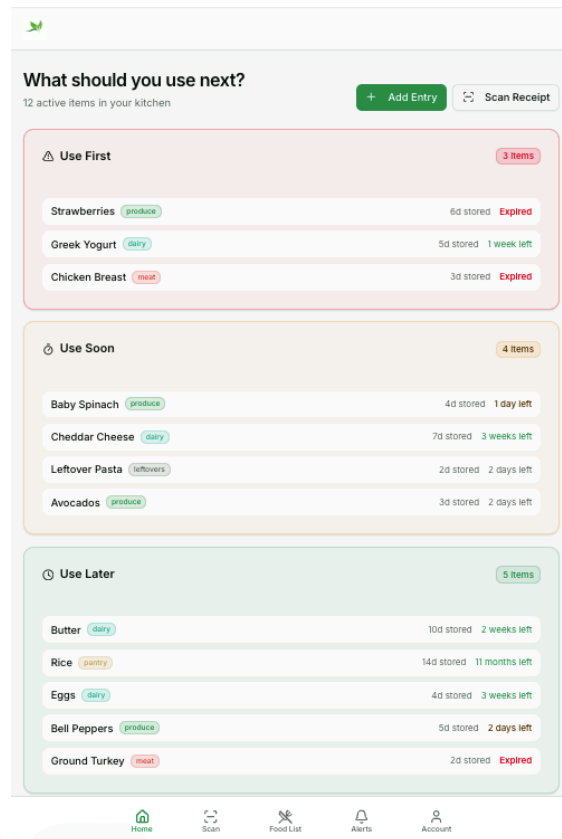


Fig 20: Paper Prototype 1

As an Active/Occasional user, I want to receive notifications so that I can stay informed about expiring items.

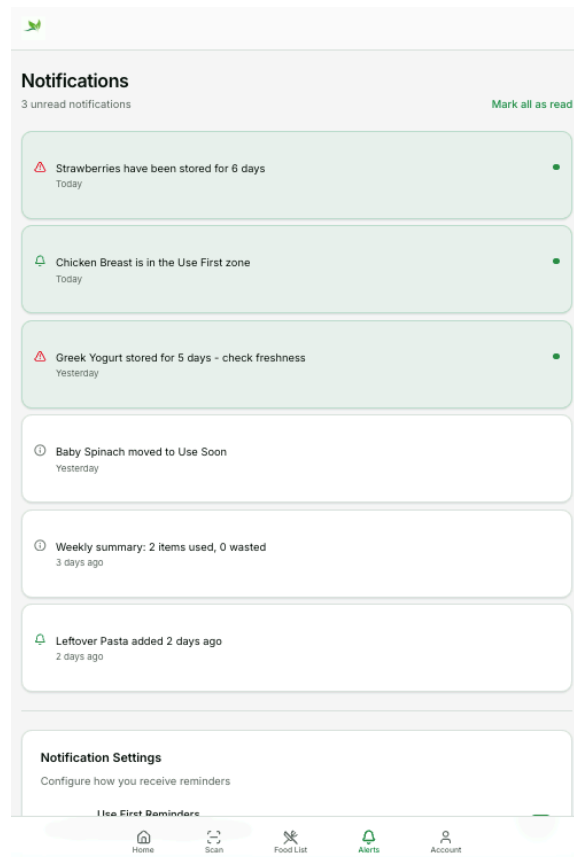


Fig 21: Paper Prototype 2

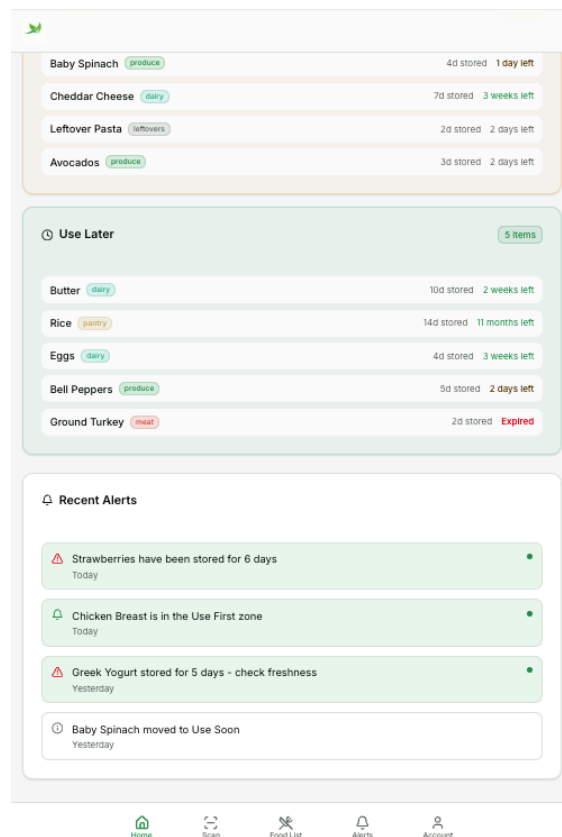


Fig 22: Paper Prototype 3

As an Active or Occasional user, I want to capture or upload a grocery receipt so that I do not have to enter each item manually.

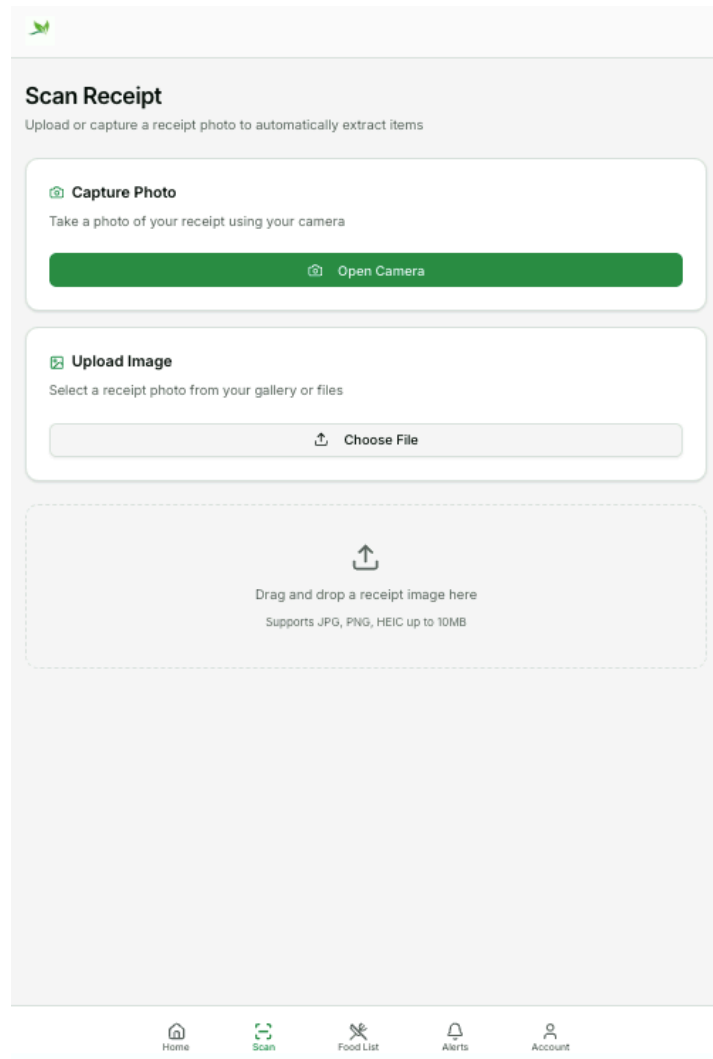


Fig 23: Paper Prototype 4

As an Active/Occasional user, I want to review items detected from the scanned receipt so that only the correct items are saved. (&) As an Active/Occasional user, I want to add and remove the items extracted from the scanned receipt so that I can remove incorrect items and add missed items. (&) As an Active/Occasional user, I want to edit the items extracted so that I can make changes to details of item such as name, category, and quantity. (&) As an Active/Occasional user, I want to save confirmed items so that they appear in my food list.

Review Extracted Items
6 items detected from your receipt. Edit anything that looks off.

[+ Add Item](#) [✓ Save Confirmed Items](#)

Organic Bananas	Produce	1 bunch	Check this	
Whole Milk	Dairy	1 gal		
Salmon Fillet	Meat	1.2 lbs	Check this	
Sourdough Bread	Pantry	1 loaf		
Mixed Berries	Produce	12 oz	Check this	
Crm Chz	Dairy	8 oz	Low confidence	

Home Scan Food List Alerts Account

Fig 24: Paper Prototype 5

As an Active or Occasional user, I want to add items manually so that I can still have items that are not on a receipt. (&) As an Active or Occasional user, I want to select a category so that items added are organized.

The image shows a paper prototype of a mobile application screen titled "Add Item". Below the title is a subtitle: "Manually add a food item to your kitchen inventory". The main content area contains a form titled "New Food Item" with a note: "Required fields are marked with *". The form has four input fields: "Name *" with a placeholder "e.g. Chicken Breast", "Category *" with a dropdown menu showing "Select a category", "Quantity (optional)" with a placeholder "e.g. 2 lbs, 1 bag", and "Photo (optional)" with a camera icon and the text "Attach photo". At the bottom of the form is a green button labeled "Save Item" with a small icon. The bottom of the screen features a navigation bar with five icons: Home, Scan, Food List, Alerts, and Account.

Fig 25: Paper Prototype 6

As an Active or Occasional user, I want to edit food items on the list so that I can make changes and keep it updated. (&) As an Active or Occasional user, I want to mark an item I have used so that I know I no longer have it. (&) As an Active or Occasional user, I want to mark an item that was wasted so that I can accurately track what was not used. (&) As an Active/Occasional user, I want to view the lifespan status of food items so that I can understand how long they have been stored.

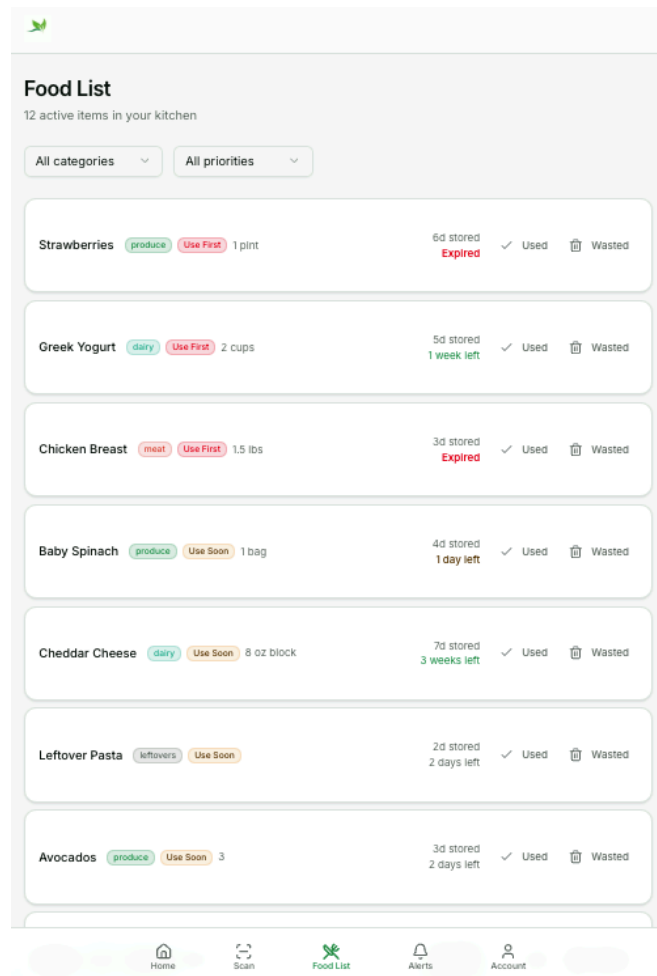


Fig 26: Paper Prototype 7

As an Admin, I want to access a system overview so that I can monitor activity across the systems.

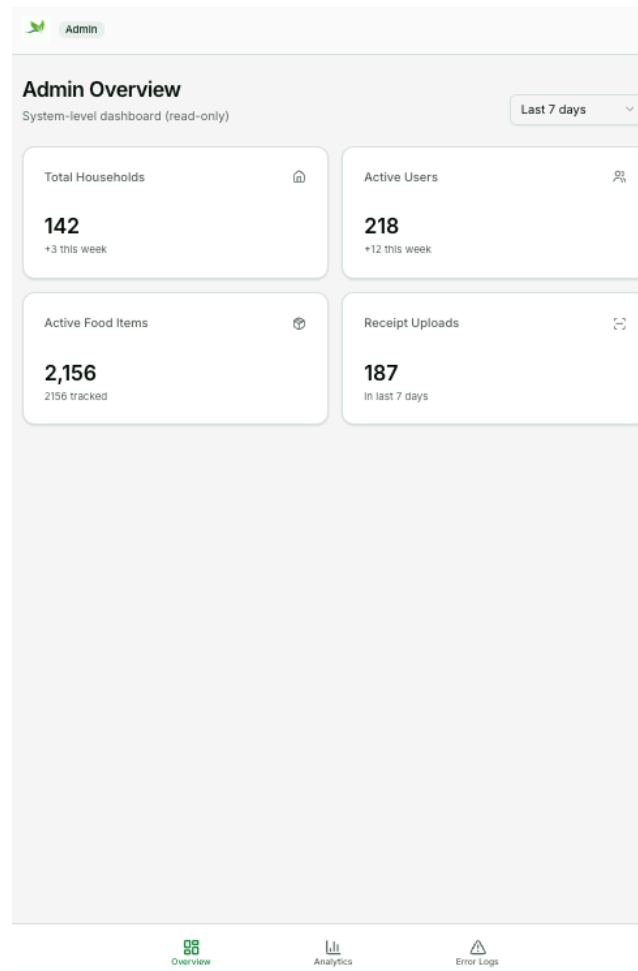


Fig 27: Paper Prototype 8

As an admin, I want to view analytics and reports so that I can monitor system usage and maintain accurate documentation.

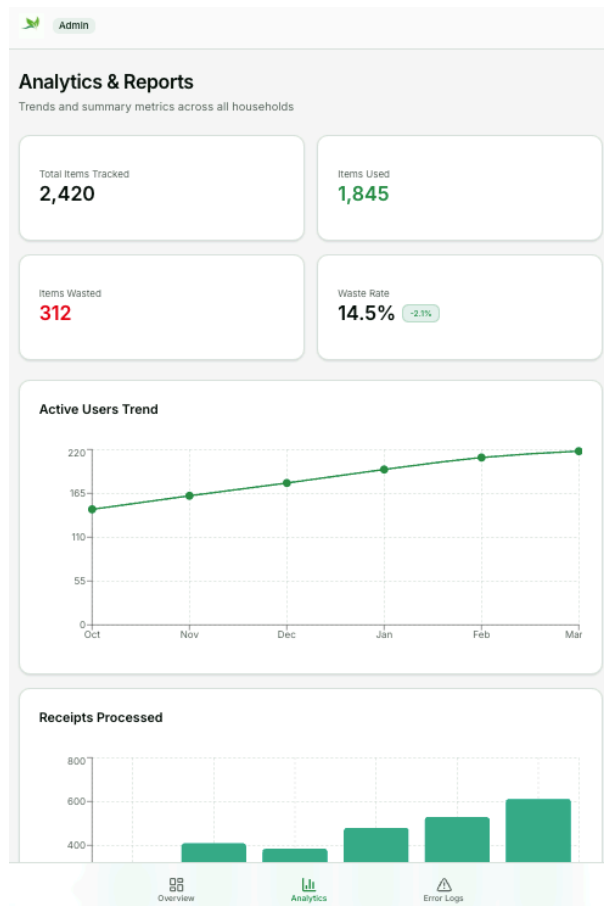


Fig 28: Paper Prototype 9

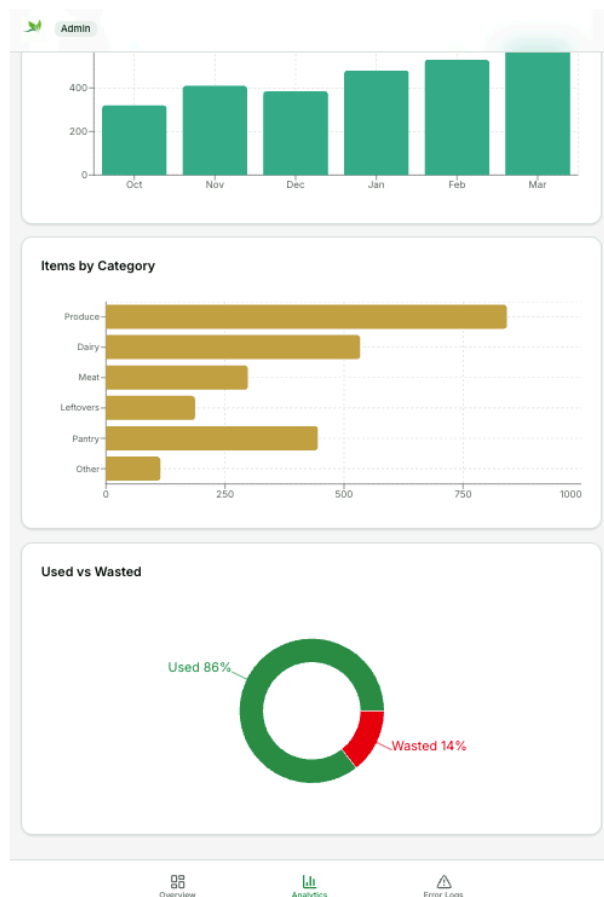


Fig 29: Paper Prototype 10

As an admin, I want to access system error logs so that I can identify and investigate issues with the system.

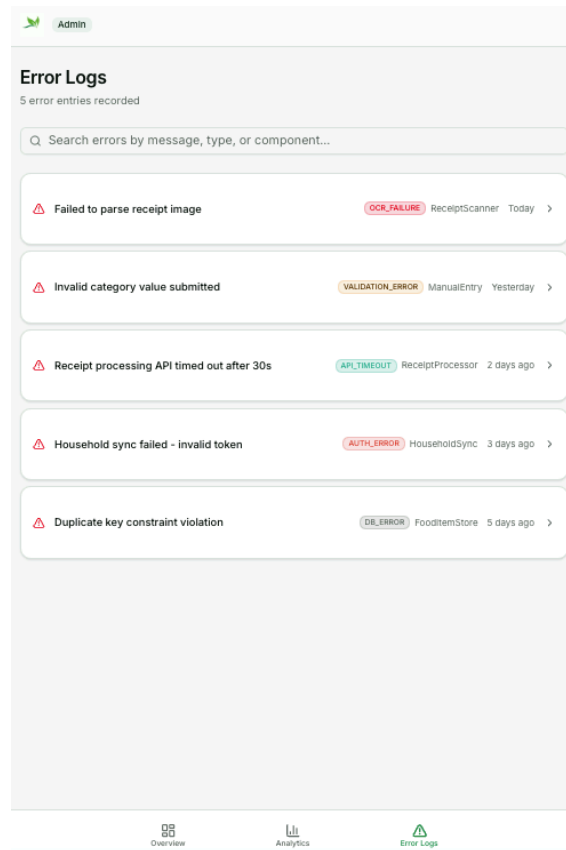


Fig 30: Paper Prototype 11